

Calibration-Aware Local Profile-Likelihood Screening in an Interacting Dark Matter-Dark Radiation Cosmology

A controlled comparison using Planck 2018, DESI DR2 BAO, and Pantheon+

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Abstract

Interacting dark matter-dark radiation (IDM-DR) can suppress the growth of cosmic structure, but the same interaction alters the cosmic microwave background (CMB). This study tested whether a deliberately small neighbourhood of IDM-DR interaction points remained competitive with an independently optimised Λ CDM baseline after correlated cosmological parameters were reoptimised. CLASS and Cobaya were used with Planck 2018 low- ℓ TT and EE, Planck high- ℓ TTTEEE-lite, DESI DR2 baryon acoustic oscillations, and Pantheon+ supernovae. Seven fixed interaction points were screened while seven shared cosmological and calibration parameters were profiled. The lowest screened boundary point, $f_{idm} = 0.0035$, $N_{idr} = 0.020$, and $a_{idm-dr} = 47612.96$, was then tightened through three additional BOBYQA trust-region stages and reevaluated in a fresh model. Its final data statistic was $\chi^2_{data} = 2426.320647$, compared with 2427.604185 for Λ CDM. Including the Planck calibration constraint gave $\Delta\chi^2_{eff} = -1.306361$. The endpoint raised H_0 by only $0.287 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and lowered S_8 by 0.003289; approximately 93% of the fractional S_8 decrease arose from the lower fitted matter density. A local one-coordinate AIC count gave $\Delta\text{AIC} = +0.693639$, while a conservative three-coordinate count gave +4.693639. The result therefore demonstrates correlated parameter compensation and local competitiveness, but it does not establish a preferred interaction, resolve either cosmological tension, or provide a global IDM-DR constraint.

Article type: Original Research

Keywords: cosmology; interacting dark matter; dark radiation; cosmic microwave background; DESI DR2; Pantheon+; profile likelihood; CLASS; Cobaya

Plain-language summary

A model can move one cosmological quantity in the desired direction while still fitting the complete dataset poorly. This study switched on an interaction between a small fraction of dark matter and dark radiation, measured the immediate change, and then allowed the usual cosmological parameters to readjust. That readjustment recovered nearly all of the original mismatch and produced a slightly lower numerical score than Λ CDM. However, the gain remained small, depended on how model complexity was counted, and occurred at the lowest tested value of one interaction coordinate. The supported conclusion is therefore not that new dark-sector physics has been discovered, but that correlated parameter adjustment can make a locally poor model appear competitive after reoptimisation.

1 Introduction

1.1 Cosmological context

The six-parameter Lambda cold dark matter model (Λ CDM) provides a compact description of the CMB, baryon acoustic oscillations (BAO), Type Ia supernova distances, and the broad growth of cosmic structure.^[1] Planck 2018 measurements interpreted within Λ CDM give H_0 near $67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, whereas a representative local distance-ladder determination gives $73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$.^[1,2] This disagreement is commonly called the Hubble tension.

A second possible disagreement concerns late-time matter clustering. Weak-lensing surveys often summarise their result using

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_m}{0.3}},$$

where σ_8 is the linear matter-fluctuation amplitude on a radius of $8h^{-1} \text{ Mpc}$ and Ω_m is the present matter-density fraction. Several analyses obtain values below the Planck-inferred prediction, although the size of the discrepancy depends on survey combinations and systematic modelling.^[3,4] A proposed extension must therefore be judged against the complete likelihood rather than by whether one derived quantity moves in a preferred direction.

1.2 Interacting dark matter and dark radiation

A richer dark sector may contain a relativistic component, called dark radiation, which exchanges momentum with a fraction of the dark matter. The ETHOS framework maps microscopic dark-sector physics onto effective parameters that govern perturbations and structure formation.^[5,6] Momentum exchange acts as a drag: modes entering the horizon while the interaction is active can grow more slowly than collisionless modes, potentially lowering σ_8 or S_8 .

The same interaction also changes gravitational potentials, acoustic driving, diffusion damping, and lensing smoothing in the CMB. Published analyses therefore find that CMB, BAO, Lyman- α , and other structure-sensitive data strongly restrict dark matter-dark radiation interactions.^[7-10] The fixed-cosmology response of an interaction and its viability after correlated parameters move are consequently different scientific questions.

1.3 Research question, hypothesis, and contribution

The research question was:

Does a small, explicitly defined neighbourhood of IDM-DR interaction points remain competitive with an independently optimised Λ CDM baseline after the shared cosmological and Planck calibration parameters are reoptimised under a modern Planck 2018 + DESI DR2 + Pantheon+ likelihood?

The physical expectation was that switching on the interaction at a common cosmology would modify growth-related quantities but produce a substantial CMB cost. The inferential expectation was that correlated changes in the standard cosmological parameters would recover part of this cost, without yielding decisive evidence for the interacting model.

The contribution is methodological rather than the proposal of a new interaction theory or a global constraint. The analysis separates: (i) the uncompensated fixed-cosmology response, (ii) recovery through correlated reoptimisation, (iii) one-coordinate local interaction sensitivity, (iv) exact endpoint reevaluation, (v) likelihood redistribution among datasets, and (vi) model-complexity sensitivity.

1.4 Scope

This is a local conditional profile-likelihood screening study. Seven selected interaction points were held fixed while seven shared cosmological and calibration parameters were reoptimised. The study is not a marginalised posterior analysis, a confidence construction, Bayesian evidence, a continuous interaction optimum, or proof of a global minimum. No weak-lensing, redshift-space-distortion (RSD), cluster-abundance, or Lyman- α likelihood was included. Reported σ_8 and S_8 values are therefore predictions under the adopted CMB+BAO+supernova likelihood, not direct measurements of improved agreement with growth data.

2 Theoretical and inferential framework

2.1 Linear response of the interacting sector

In linear perturbation theory, interacting dark matter and dark radiation exchange momentum. Schematically, the interacting-dark-matter velocity divergence contains a drag contribution proportional to $\Gamma_{idm-dr}(\theta_{dr} - \theta_{idm})$, while the dark-radiation hierarchy contains the opposite exchange. The detailed coefficients depend on the implementation and whether the dark radiation is fluid-like or free-streaming. The drag opposes relative motion and can delay growth of the interacting component.

2.2 Matter clustering and S_8

The linear matter power spectrum $P(k, z)$ determines the smoothed variance

$$\sigma_R^2(z) = \frac{1}{2\pi^2} \int_0^\infty k^2 P(k, z) W^2(kR) dk.$$

Setting $R = 8h^{-1}$ Mpc defines σ_8 . Because S_8 combines σ_8 and Ω_m , a lower fitted S_8 can result from a lower fluctuation amplitude, a lower matter density, or both. Profiling can therefore change the physical reason for an apparently favourable movement.

2.3 Conditional profile likelihood

For fixed interaction coordinates ϕ and profiled shared parameters θ ,

$$\chi_{prof}^2(\phi) = \min_{\theta} \chi^2(\theta, \phi).$$

A profile retains the lowest score located at each fixed interaction point; it does not integrate over parameter-space volume.^[11] Because only seven interaction points were screened and the optimisation was local, the manuscript uses the phrase *lowest tested endpoint* rather than *measured best-fit interaction*.

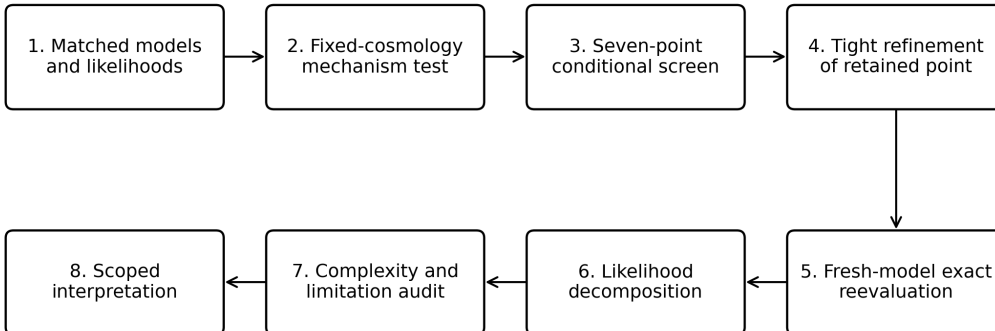


Figure 1. Computational and inferential workflow. Each stage answers a distinct question, from the uncompensated response to the scoped model-complexity interpretation.

3 Methods

3.1 Software environment and source provenance

CLASS calculated the background evolution, perturbations, CMB spectra, linear matter power spectrum, and derived quantities.^[12] Cobaya supplied the parameterisation, likelihood interfaces, and optimisation framework.^[13] The validated environment used Python 3.12, Cobaya 3.6.2, and Py-BOBYQA 1.5.0 in Google Colab, with configurations and outputs stored on Google Drive.

The custom CLASS source directory did not retain its upstream Git metadata. A recursive source-tree SHA-256 checksum was recorded as 05264dfbddd18970dcc0cf6b6d6a918749a20032af64c09bf7ce7361e926cabf. This preserves file-level identity but is weaker provenance than an upstream commit hash. The public repository documents the required interaction options and this limitation.

3.2 Observational likelihood

The combined likelihood contained Planck 2018 low- ℓ TT, Planck 2018 low- ℓ EE, Planck high- ℓ Plik TTTEEE-lite-native, DESI DR2 BAO, and Pantheon+ supernovae.^[14–16]

Likelihood	Principal sensitivity	Role in this study
Planck low- ℓ TT	Large-angle temperature anisotropy	Tests the largest-scale temperature response
Planck low- ℓ EE	Reionisation-scale polarisation	Constrains τ and its covariance with A_s
Planck high- ℓ TTTEEE-lite	Acoustic peaks, damping, polarisation, and lensing smoothing	Provides the most detailed early-Universe spectral test
DESI DR2 BAO	Distances relative to r_{drag}	Tests late-time geometry and sound-ruler consistency
Pantheon+	Relative luminosity distances	Constrains the shape of the expansion history

Table 1. Observational information entering the modern combined likelihood.

3.3 Model specification and profiled parameters

The independently constructed Λ CDM baseline omitted all IDM-DR inputs. The central interaction point fixed $f_{\text{idm}} = 0.0035$, $N_{\text{idr}} = 0.033$, $a_{\text{idm-dr}} = 47612.96$, $n_{\text{index,idm-dr}} = 4$, and free-streaming dark radiation. Six neighbours varied one interaction coordinate at a time. The numerical normalisation of $a_{\text{idm-dr}}$ is specific to this CLASS implementation and should not be compared directly with differently normalised interaction parameters.

The neutrino sector was fixed to one massive species with $m_\nu = 0.06$ eV, $N_{\text{ncdm}} = 1$, and $N_{\text{ur}} = 2.0328$. At each interaction point, the profiled vector was $(\omega_b, \omega_{\text{cdm}}, H_0, \tau_{\text{reio}}, \ln(10^{10} A_s), n_s, A_{\text{Planck}})$.

Parameter	Lower bound	Upper bound	Treatment
ω_b	0.0200	0.0245	Uniform bounded search
ω_{cdm}	0.1000	0.1400	Uniform bounded search

Parameter	Lower bound	Upper bound	Treatment
H_0 [km s ⁻¹ Mpc ⁻¹]	55.0	85.0	Uniform bounded search
τ_{reio}	0.010	0.120	Uniform bounded search
$\ln(10^{10} A_s)$	2.80	3.30	Uniform bounded search
n_s	0.900	1.050	Uniform bounded search
A_{Planck}	-	-	Gaussian $\mathcal{N}(1, 0.0025^2)$

Table 2. Profiled shared parameters and numerical ranges.

Label	f_{idm}	N_{idr}	a_{idm-dr}
central	0.0035	0.033	47612.96
f_{low}	0.0020	0.033	47612.96
f_{high}	0.0050	0.033	47612.96
N_{low}	0.0035	0.020	47612.96
N_{high}	0.0035	0.050	47612.96
a_{low}	0.0035	0.033	30000
a_{high}	0.0035	0.033	70000

Table 3. Seven fixed interaction points in the local conditional screen.

3.4 Optimisation and exact endpoint validation

The Λ CDM baseline used four separated BOBYQA starts followed by progressively tighter trust-region radii.^[17,18] Its final saved point was reevaluated directly in a fresh model. The central IDM-DR run started from the Λ CDM cosmology. Each neighbour began from the central profiled coordinates, used a 350-evaluation cap and $\rho_{end} = 0.05$, and was independently reevaluated.

The lowest screening endpoint, $N_{idr} = 0.020$, was then continued through $\rho_{end} = 0.005$, 0.001, and 0.0002. At every stage, the retained point was evaluated in a newly constructed CLASS-Cobaya model. An endpoint was accepted only when the optimiser returned a complete state, all five likelihood components were finite, and the machine-readable result could be traced to its source run.

3.5 Calibration and model-complexity diagnostics

The primary statistic was

$$\chi_{data}^2 = -2 \sum_i \ln \mathcal{L}_i.$$

The Planck calibration parameter carried the Gaussian contribution

$$\chi_{cal}^2 = \left(\frac{A_{Planck} - 1}{0.0025} \right)^2,$$

and the calibration-aware effective statistic was $\chi_{eff}^2 = \chi_{data}^2 + \chi_{cal}^2$.

The Akaike Information Criterion was calculated as $AIC = \chi_{eff}^2 + 2k$.^[19] Three interpretations were shown: an equal-dimension fixed-endpoint comparison, a local one-coordinate count, and a conservative

count treating all three interaction coordinates as additional degrees of freedom. These are sensitivity diagnostics rather than formal global model-selection results.

3.6 Reporting rules for a small endpoint difference

The analysis reports the sign and size of the endpoint difference but does not translate it into a p -value, Gaussian significance, confidence interval, or posterior probability. The phrase *locally competitive* means only that the retained endpoint has a similar or slightly lower numerical objective under the tested likelihood. It does not imply that IDM-DR is required by the data.

4 Results

4.1 Validated Λ CDM baseline and fixed-response penalty

The exact modern Λ CDM endpoint gave $\chi_{data}^2 = 2427.604184715953$, $H_0 = 68.220876 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.303141067$, $\sigma_8 = 0.815965320$, and $S_8 = 0.820225866$.

At the optimised Λ CDM cosmology, switching on the central IDM-DR interaction without allowing the shared parameters to move gave $\chi_{data}^2 = 2444.283891763638$, an immediate penalty of +16.679707. After reoptimising the seven shared parameters at the central interaction point, the exact statistic fell to 2427.051083129154, a raw gain of 0.553102 relative to Λ CDM. Parameter compensation therefore recovered the full fixed-point penalty and slightly exceeded the baseline score.

Analysis stage	Fixed interaction coordinates	χ_{data}^2	Difference from Λ CDM
Λ CDM exact endpoint	None	2427.604185	0.000000
Central IDM-DR at Λ CDM cosmology	central point	2444.283892	+16.679707
Central IDM-DR after profiling	central point	2427.051083	-0.553102
N_{low} coarse screen	N_{low} point	2426.675712	-0.928473
N_{low} tight refinement	N_{low} point	2426.320647	-1.283538

Table 4. Separation of the uncompensated interaction response, profiled recovery, local screening, and tight refinement.

4.2 Seven-point local screen and tight refinement

The six one-coordinate neighbours and the central point formed a coarse local screen. The lowest screened endpoint was N_{low} , at the lowest tested value $N_{idr} = 0.020$. Three tighter continuation stages reduced its exact statistic from 2426.675712 to 2426.320647. The final raw gain relative to Λ CDM was therefore 1.283538.

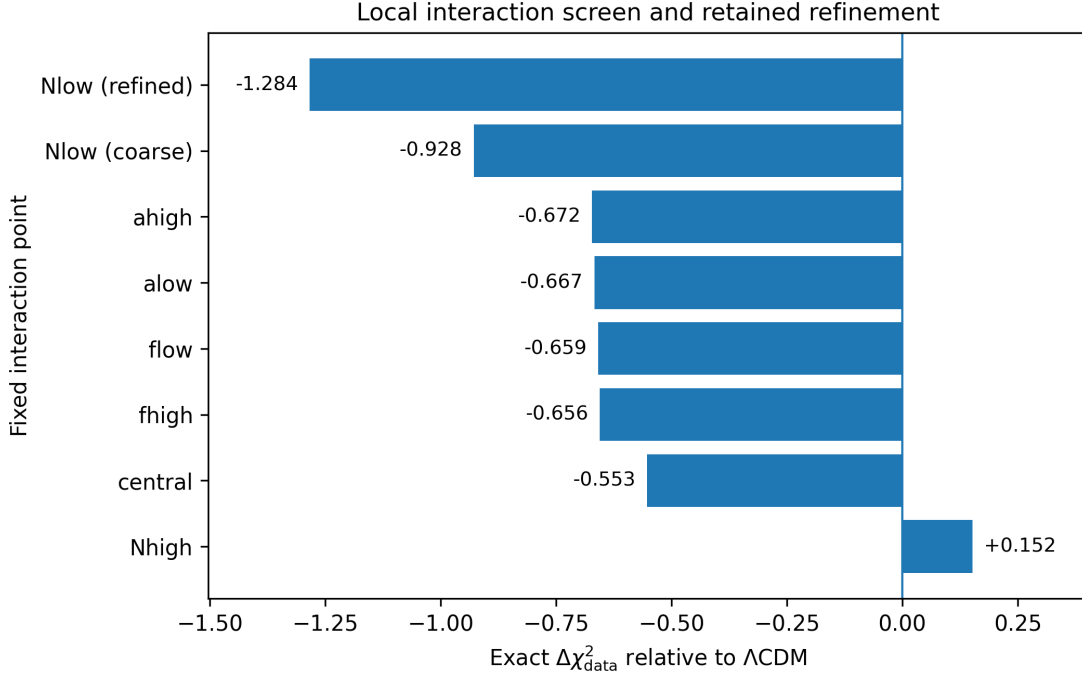


Figure 2. Exact data-statistic differences for the seven coarse screen points and the tightly refined retained endpoint. Negative values are lower than the independently evaluated ΛCDM baseline.

Point	f_{idm}	N_{idr}	$a_{\text{idm-dr}}$	Exact χ^2_{data}	$\Delta\chi^2$ vs ΛCDM
N_{low} , refined	0.0035	0.020	47612.96	2426.320647	-1.283538
N_{low} , coarse	0.0035	0.020	47612.96	2426.675712	-0.928473
a_{high}	0.0035	0.033	70000.00	2426.931795	-0.672390
a_{low}	0.0035	0.033	30000.00	2426.937004	-0.667180
f_{low}	0.0020	0.033	47612.96	2426.945523	-0.658662
f_{high}	0.0050	0.033	47612.96	2426.948433	-0.655752
central	0.0035	0.033	47612.96	2427.051083	-0.553102
N_{high}	0.0035	0.050	47612.96	2427.755751	+0.151566

Table 5. Complete local sensitivity screen with the retained endpoint after tight refinement.

4.3 Boundary behaviour in N_{idr}

The N_{idr} direction changed more strongly than the f_{idm} and $a_{\text{idm-dr}}$ neighbours. However, $N_{\text{idr}} = 0.020$ was the lowest value tested. The analysis therefore locates a boundary endpoint, not an interior optimum. It does not determine a preferred N_{idr} value, confidence interval, or lower bound.

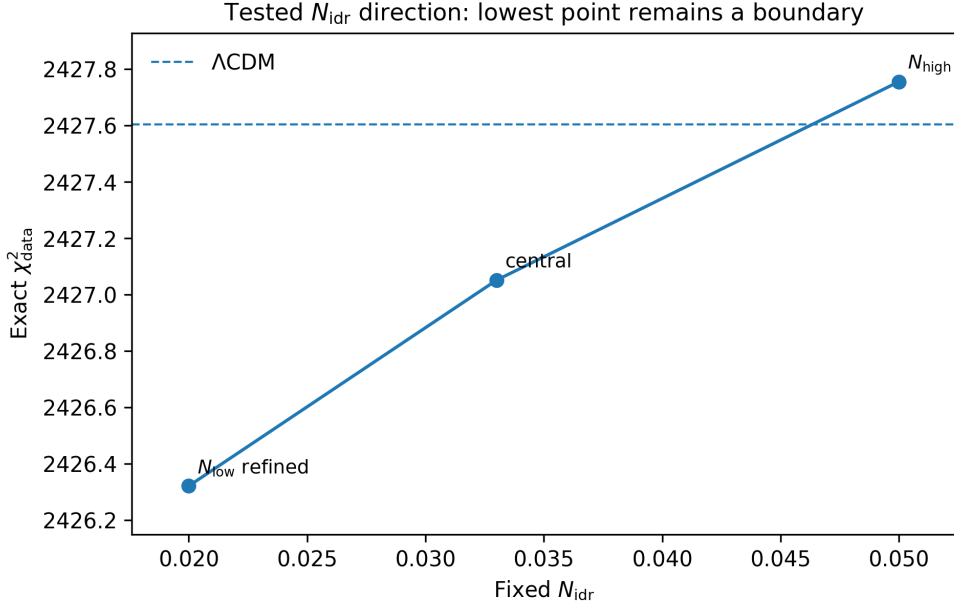


Figure 3. Exact χ^2_{data} across the three tested N_{idr} values, with the lower point tightly refined. The lowest point remains at the lower boundary of the tested grid.

4.4 Refinement history and numerical stability

The exact refinement sequence was 2426.675712, 2426.546777, 2426.378541, and 2426.320647. The final tightening changed the statistic by 0.057894, about 4.5% of the total raw gain relative to ΛCDM . The declining final increment is consistent with local numerical stabilisation along this continuation path, but it is not proof of global convergence or independence from the starting coordinates.

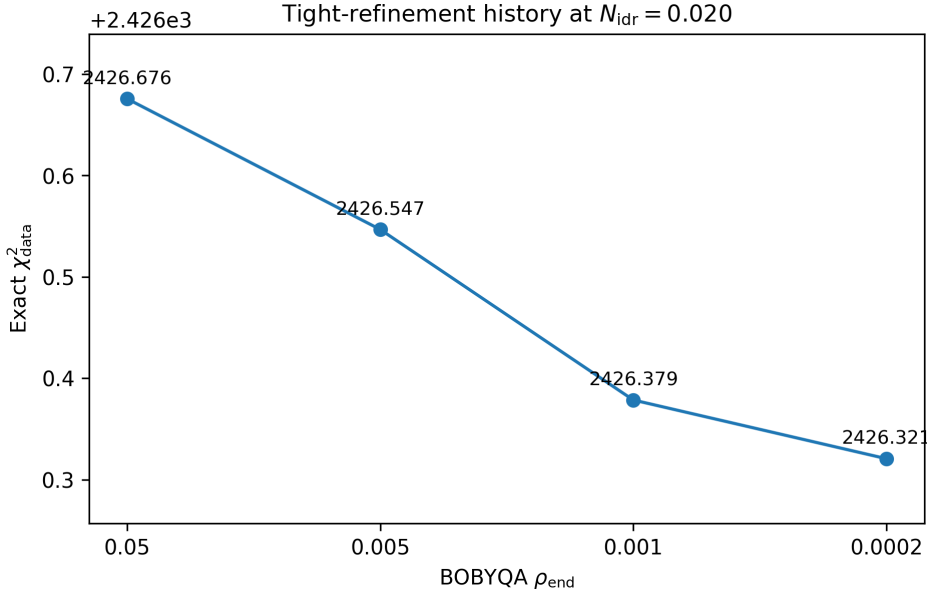


Figure 4. Exact tight-refinement history at the retained boundary endpoint.

Stage	ρ_{end}	Exact χ^2_{data}	Gain vs ΛCDM
Coarse screen	0.05	2426.675712	0.928473
Refinement 1	0.005	2426.546777	1.057407
Refinement 2	0.001	2426.378541	1.225644
Refinement 3	0.0002	2426.320647	1.283538

Table 6. Exact refinement history for $N_{idr} = 0.020$.

4.5 Derived-parameter shifts

Relative to Λ CDM, the refined endpoint changed H_0 by $+0.287216 \text{ km s}^{-1} \text{ Mpc}^{-1}$, Ω_m by -0.002253409 , σ_8 by -0.000234110 , and S_8 by -0.003288730 . A logarithmic decomposition attributes approximately 93% of the fractional S_8 decrease to the lower Ω_m and about 7% to the lower σ_8 . The movement is therefore mainly a density effect rather than direct evidence of sustained interaction-driven growth suppression.

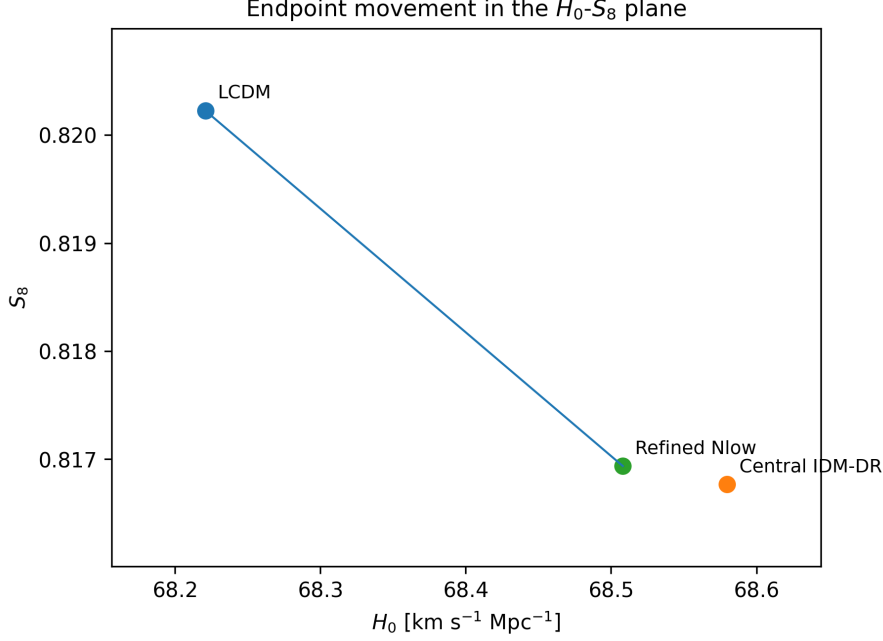


Figure 5. Movement from the exact Λ CDM endpoint to the central and tightly refined IDM-DR endpoints in the H_0 - S_8 plane. The line indicates endpoint displacement, not a confidence contour or sampled trajectory.

Quantity	Λ CDM	Refined IDM-DR	Difference
χ^2_{data}	2427.604185	2426.320647	-1.283538
Calibration penalty	0.024594	0.001771	-0.022823
χ^2_{eff}	2427.628778	2426.322418	-1.306361
H_0 [$\text{km s}^{-1} \text{ Mpc}^{-1}$]	68.220876	68.508092	+0.287216
Ω_m	0.303141	0.300888	-0.002253
σ_8	0.815965	0.815731	-0.000234
S_8	0.820226	0.816937	-0.003289
r_{drag} [Mpc]	147.537980	147.334417	-0.203563
Age [Gyr]	13.772990	13.743911	-0.029079

Table 7. Exact endpoint comparison. Differences are IDM-DR minus Λ CDM.

4.6 Likelihood redistribution

The final endpoint was reevaluated with all five likelihood components extracted exactly. Relative to Λ CDM, the refined endpoint improves Planck low- ℓ TT, Planck low- ℓ EE, and DESI DR2 BAO, while worsening Planck high- ℓ TTTEEE-lite and Pantheon+. The lower total is therefore a cross-dataset compromise rather than a uniform improvement.

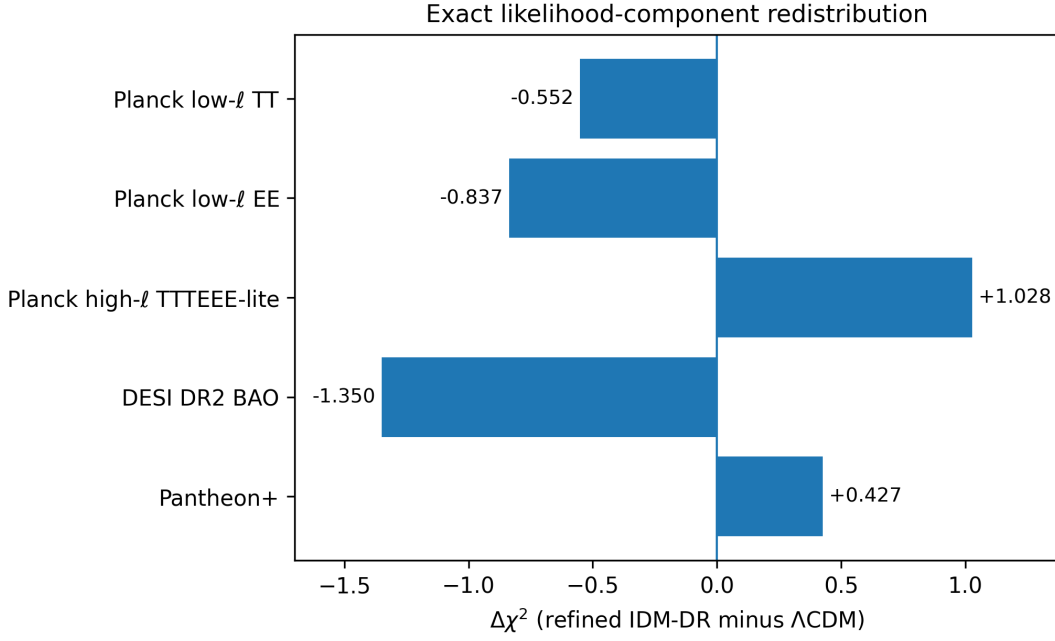


Figure 6. Exact likelihood-component changes at the tightly refined endpoint. Negative values improve the component score.

Component	Λ CDM χ^2	Refined IDM-DR χ^2	Difference
Planck low- ℓ TT	23.250957	22.699083	-0.551873
Planck low- ℓ EE	402.718334	401.881574	-0.836760
Planck high- ℓ TTTEE-lite	583.060962	584.089305	+1.028344
DESI DR2 BAO	13.135413	11.785622	-1.349792
Pantheon+	1405.438519	1405.865062	+0.426544
Total	2427.604185	2426.320647	-1.283538

Table 8. Exact likelihood-component redistribution at the tightly refined endpoint.

4.7 Complexity-adjusted comparison

Using the exact calibration coordinates gives $\Delta\chi_{eff}^2 = -1.306361$. The equal-dimension endpoint diagnostic therefore favours the retained numerical endpoint. Adding one local interaction degree of freedom gives $\Delta\text{AIC} = +0.693639$, which indicates no meaningful AIC distinction. Counting all three interaction coordinates gives $\Delta\text{AIC} = +4.693639$, favouring the simpler baseline.

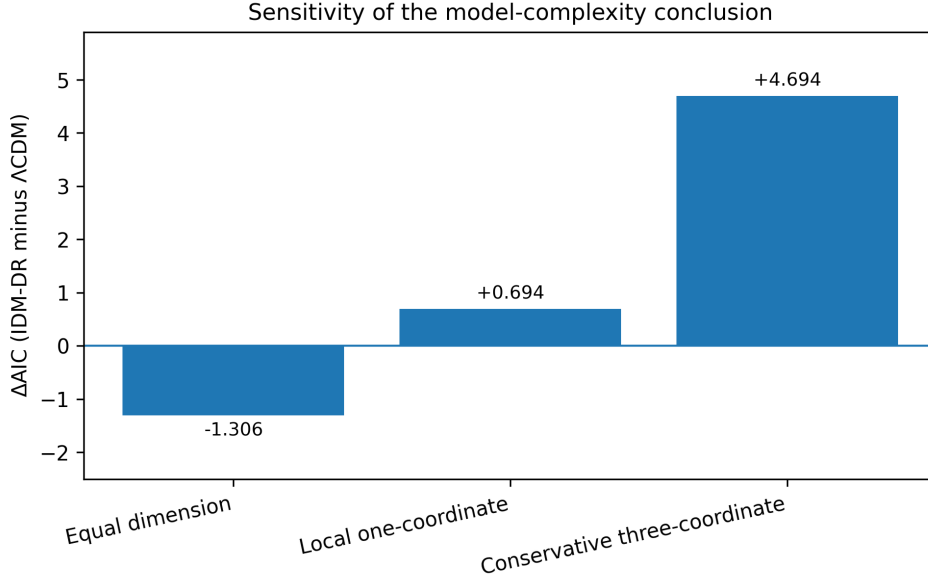


Figure 7. Sensitivity of the model-complexity conclusion. Positive ΔAIC values favour Λ CDM.

Interpretation	$k_{\Lambda\text{CDM}}$	$k_{\text{IDM-DR}}$	ΔAIC
Equal-dimension fixed endpoints	7	7	-1.306361
Local one-coordinate count	7	8	+0.693639
Conservative three-coordinate extension	7	10	+4.693639

Table 9. Exact AIC sensitivity to parameter counting. Positive values favour Λ CDM.

5 Discussion

5.1 Fixed response and profiled viability answer different questions

The central interaction was strongly disfavoured when switched on at the Λ CDM cosmology, yet became locally competitive after the shared parameters were reoptimised. This directly demonstrates cosmological compensation: an uncompensated spectral mismatch can lie partly along parameter-degeneracy directions. The recovery is methodologically informative, but it is not evidence that the interaction is required.

5.2 The retained S_8 shift is mainly a density shift

Both σ_8 and Ω_m are lower at the refined endpoint, but the matter-density change supplies about 93% of the logarithmic S_8 decrease. The final movement is therefore not a pure measure of drag-induced growth suppression. Interpreting only the summary parameter would obscure the mechanism by which profiling produced the change.

5.3 Dataset compromise

DESI DR2 BAO and the low- ℓ CMB components improve, whereas the high- ℓ CMB and Pantheon+ terms worsen. The endpoint is therefore a compromise across the combined likelihood. The remaining

high- ℓ cost is consistent with the known sensitivity of precision CMB spectra to interacting dark-sector perturbations.^[7-10]

5.4 Implications for cosmological tensions

The H_0 increase of approximately $0.29 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is far smaller than the several- $\text{km s}^{-1} \text{ Mpc}^{-1}$ separation between representative CMB and local distance-ladder values. The tested neighbourhood does not resolve the Hubble tension. The S_8 reduction is also modest and was not tested against a direct growth likelihood, so it cannot be described as improved agreement with DES, KiDS, RSD, cluster, or Lyman- α measurements.

5.5 Numerical robustness and unresolved uncertainty

Tight refinement increased the raw endpoint gain from 0.928 to 1.284. The final trust-region stage still changed the exact statistic by 0.057894. This is small compared with the total gain but not negligible enough to claim convergence at the 0.01 level. More importantly, the retained endpoint was followed along one continuation path. An independent-start repeat, a denser lower- N_{idr} scan, and a local Hessian or posterior calculation would be required to quantify robustness and uncertainty.

5.6 Strengths

The main strengths are the independently optimised Λ CDM baseline, matched likelihood definitions, separation of fixed-response and reoptimised stages, transparent reporting of every interaction point, three-stage tight refinement, exact fresh-model reevaluation, exact component-level auditing, machine-readable results, public code and data mapping, and deliberately conservative interpretation.

5.7 Limitations and consequences

Limitation	Consequence
Lowest N_{idr} point lies at a tested boundary	No interior optimum, confidence interval, or preferred value can be claimed
One continuation path for the retained endpoint	Dependence on starting coordinates cannot be excluded
Interaction coordinates were not jointly optimised	No global IDM-DR constraint or formal likelihood-ratio result
No posterior or Hessian uncertainty region	Endpoint shifts cannot be compared with fitted error bars
No direct growth likelihood	Lower S_8 is not evidence of improved weak-lensing or RSD agreement
Final tightening still changed χ^2 by 0.057894	Global numerical convergence is not demonstrated
Missing upstream CLASS Git commit	Reproduction relies on a source-tree checksum and documented build compatibility

Table 10. Principal limitations and their effect on interpretation.

The highest-priority extensions are an independent-start repeat of the refined endpoint, additional values below $N_{idr} = 0.020$, and a dense or sampled joint exploration of the interaction coordinates. A converged MCMC or validated nested-sampling analysis would be required before posterior constraints or evidence-based model comparison could be reported.

6 Conclusion

This study compared a modern Planck 2018 + DESI DR2 + Pantheon+ Λ CDM baseline with a seven-point local IDM-DR screen and tight refinement of the lowest tested boundary endpoint. Switching

on the central interaction at the Λ CDM cosmology produced a +16.679707 data-statistic penalty. Reoptimisation recovered this mismatch, and the final refined endpoint at $f_{idm} = 0.0035$, $N_{idr} = 0.020$, and $a_{idm-dr} = 47612.96$ gave $\chi^2_{data} = 2426.320647$, compared with 2427.604185 for Λ CDM.

The calibration-aware difference was $\Delta\chi^2_{eff} = -1.306361$. The endpoint raised H_0 by only $0.287 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and lowered S_8 by 0.003289, mainly through a lower matter density. Neither shift resolves the associated cosmological tension. The local one-coordinate AIC result was nearly neutral, while the conservative three-coordinate count favoured Λ CDM. Because the retained point is a grid boundary and was refined along one continuation path, the supported conclusion is narrow: correlated parameter compensation makes the tested IDM-DR neighbourhood locally competitive, but the present calculation does not establish model preference, a global optimum, or evidence for new dark-sector physics.

Declarations

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Competing interests

The author declares no competing interests.

Ethics statement

This computational study used public cosmological likelihoods and software. It involved no human participants, animal subjects, clinical records, or personally identifiable information.

Author contributions

Parv Gidiya formulated the computational question, assembled and ran the analysis, audited the numerical outputs, interpreted the results, prepared the figures and tables, and drafted the manuscript.

Data and code availability

The exact configurations, machine-readable results, refinement history, component tables, figure-generation scripts, validation tools, and environment records are publicly available at <https://github.com/parvgidiya/idmdr-modern-profile-likelihood>. Version 1.0.0 is permanently archived on Zenodo at <https://doi.org/10.5281/zenodo.21741405>.^[20] Third-party likelihood datasets are not redistributed. A compatible custom CLASS source tree is required for independent reruns; its recorded source-tree checksum and build-provenance note are included.

Computational and language-assistance disclosure

Computational assistance tools were used for code debugging, file organisation, figure preparation, and language editing. Every numerical result in the manuscript was taken from the author's executed analysis output or independently recomputed from that output. The author reviewed the interpretation and accepts responsibility for the scientific content.

Author biography

Parv Gidiya is a 13-year-old student at DPS Monarch International School in Doha, Qatar. He is interested in astronomy, cosmology, and computational physics, particularly the use of public scientific software and observational data to test extensions of the standard cosmological model.

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Appendix A. Exact endpoint coordinates

The sampled coordinates below are the exact records used for the canonical comparison.

Parameter	Λ CDM	Refined IDM-DR
ω_b	0.022456882	0.022536034
ω_{cdm}	0.117983500	0.118037200
H_0	68.220876	68.508092
τ_{reio}	0.071294617	0.070244317
$\ln(10^{10} A_s)$	3.071522200	3.070654500
n_s	0.968697530	0.971546570
A_{Planck}	0.999607940	1.000105200

Table A1. Sampled endpoint coordinates used in the final exact comparison.

Appendix B. Reproducibility contents

- Exact Λ CDM and seven IDM-DR YAML configurations
- Final refined endpoint JSON and CSV records
- Raw continuation-profile output and exact component tables
- Figure source tables and rendered figures
- Scripts for minimisation, exact reevaluation, validation, and optional boundary extension
- Pinned package versions and environment metadata
- CLASS source-tree checksum and provenance note
- README, reproduction guide, citation metadata, and results map

Appendix C. Claim-evidence map

Manuscript claim	Machine-readable source
Exact Λ CDM endpoint	<code>results/lcdm/lcdm_exact_summary.json</code>
Complete seven-point ordering	<code>results/derived/modern_idmdr_seven_point_profile.csv</code>
Final refined endpoint	<code>results/idmdr/final/modern_idmdr_final_best_endpoint.json</code>
Exact component redistribution	<code>results/idmdr/final/modern_idmdr_final_best_chi2_components.csv</code>
AIC sensitivity	<code>results/idmdr/final/modern_idmdr_final_best_endpoint.json</code>
Original completed run record	<code>results/raw/neighbor_profile_stdout.txt</code>

Table C1. Direct mapping from central claims to repository evidence.